

SATYA SRUJANA PILLI

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EDUCATION

University of Massachusetts Amherst

MS IN COMPUTER SCIENCE

Amherst, MA | SEP 2024 – MAY 2026

CGPA: 3.96

Birla Institute of Technology and Science, Pilani

B.E. COMPUTER SCIENCE

Hyderabad, IN | AUG 2018 – JUN 2022

CGPA: 8.33

SKILLS

Languages: Java, C++, Python, C, Ruby, Scala, MySQL, JavaScript, HTML, CSS, Shell, MATLAB

Frameworks: TensorFlow, PyTorch, scikit-learn, Keras, OpenCV, SpringBoot, REST API, Web services, MERN, ReactJS

Technologies: Git, AWS, Docker, WebLabs, Kubernetes, Helm, Postman, Postgres, MongoDB, DynamoDB, CloudWatch

AI Technologies: OpenAI, Gemini, Claude APIs, Mistral, RAG, MCP, ONNX, Cursor, GitHub Copilot

Soft Skills: Cross-functional Collaboration, Technical Writing, Stakeholder Communication, Agile/Scrum

WORK EXPERIENCE

GRADUATE RESEARCH EXTERN | Cisco

San Francisco, CA | JAN 2026 - MAY 2026

- Built an **adaptive LLM monitoring agent** with **ReAct-style** reasoning to detect sabotage, task drift, and hidden objectives across **50+ autonomous agent trajectories**.
- Devised multi-step inspection strategies—tool-call analysis, hypothesis generation, and belief updates—improving **threat detection recall by 30%** over static classifiers.
- Benchmarked **static monitoring approaches** against adaptive reasoning-based monitors via ablation studies, quantifying recall and precision trade-offs.

GRADUATE RESEARCH ASSISTANT | CIIR LAB

Amherst, MA | APR 2025 – SEP 2025

- Embedded **gaze-based preference** signals into the training pipeline, cutting multi-session data mismatches by **40%** across **200+ user sessions**.
- Redesigned evaluation tables for **5+ domain-specific** gaze and query inputs, boosting evaluation reliability by **30%** and halving experiment cycle time.
- Hardened preference storage and logging logic, improving **reward-model accuracy by 25%** and eliminating silent data-loss bugs.

AMAZON | SDE-1

Bangalore, IN | JUL 2022 – MAR 2024

- Spearheaded design and launch of **credit-based installment plans** across **6 EU and UK markets**, boosting customer UX by **20%** through seamless payment workflows.
- Engineered item-level payment controls for **Item-Level Financing (ILF)**, increasing **customer engagement by 28%** with flexible, personalized payment methods.
- Optimized **full-stack payment flows** for Amazon Payments, cutting transaction latency by **35%** and driving **customer retention up by 15%**.
- Drove **A/B experimentation** via WebLabs across **12 feature launches**, achieving a **99.9%** deployment success rate in CI/CD pipelines.

AMAZON | SDE-INTERN

Bangalore, IN | JAN 2022 – JUN 2022

- Developed **Vapor**, an internal planning and analytics tool, streamlining project tracking for **50+ teams** and reducing planning overhead by **20%** upon production rollout.
- Implemented **service profilers** for performance monitoring, ensuring **100% bug-free deployment** across **10+ interconnected services**.
- Integrated **consistency checks** to validate transactional data, cutting error rates by **25%** and boosting **data integrity** across **3 distributed systems**.

PROJECTS

LLM-POWERED CRIME ELEMENT ANALYZER

PYTHON 3, FLASK, OLLAMA MISTRAL-7B

UNDER PROF. BRIAN LEVINE, UMASS RESCUELAB

- Designed a forensic analysis pipeline using **Mistral 7B** and **Gemma 3** to extract crime elements from chat data via **few-shot prompting**, cutting manual review time by **60%**.
- Deployed a production-ready **Flask-ML API** with CLI for batch processing and multi-format input (CSV, PDF, XLSX), serving **3 active investigations** across **10,000+ messages**.

BRIEFLY

REACTJS, PYTHON, FLASK, MISTRALAI API, MONGODB, AXIOS, HTML, CSS

- Launched a summarization platform achieving **95% accuracy** via MistralAI API, processing **100+ text files** daily.
- Orchestrated backend processing with Python and Flask, cutting processing time by **30%** and delivering **real-time** summaries to **50+ daily users**.